Assignment #2

The Invisible Patient: Reconstructing an Image

Topics 1–9 | 25 marks | Linear Algebra and Computed Tomography

The Challenge

Imagine trying to discover what is inside a sealed box without opening it. You can shine X-rays through it from different directions, but your detector only tells you how much of each beam made it through. A single measurement cannot tell you exactly *where* the X-rays were absorbed. However, measurements along several paths may allow you to reconstruct the hidden object. This is the central idea behind *computed tomography* (CT), which uses many X-ray measurements and a computer to form cross-sectional images.

In this assignment, your "patient" is only a tiny grid of four squares (pixels). Each square represents a region of material that weakens an X-ray beam by some unknown amount. Your job is to recover those four values using linear algebra, then investigate why the same idea becomes difficult in real medical imaging.

In this assignment, you will build an *idealized* two-dimensional X-ray reconstruction model using vectors, dot products, projections, matrices, reduced row echelon form (RREF), and NumPy. You will then research a limitation of real computed tomography (CT) and design a small computational investigation.

Submission. Submit **one Jupyter notebook (`.ipynb`) to LEARN**, containing explanations, mathematical working, executable Python, outputs, figures, and references. You may work individually or in a group of up to four students. Show your reasoning, not only final answers. For every part, include a brief **"What does this mean?"** explanation connecting your calculation to the imaging problem. Explanations must refer to your own numerical results, not generic descriptions. Hand calculations may be clearly typed or inserted as legible images. All group members must contribute and be identified in the submission.

Grading

Component	Marks
A. Describe the hidden image and measurements	3
B. Dot products and projection	4
C. Construct and solve the linear system	5
D. Reconstruct and verify with Python	4
E. Missing information and RREF	3
F. Research extension: when reconstruction goes wrong	6
Total	**25**

Computational Pathways and Academic Integrity

AI-assisted pathway. Generative AI is permitted but not required. You may use it to investigate unfamiliar physics, understand documentation, develop or debug code, or explore your extension. Identify each AI platform used and include/export your **complete relevant AI conversation(s)**, including prompts and responses, with your submission. Verify all mathematical claims and literature references independently.

Independent pathway. You may complete every part without AI using course notes, Python documentation, textbooks, and scientific literature. State that you did not use generative AI. Cite any external material, equations, datasets, or code you use.

Both pathways: Include clear code comments, distinguish your own work from sourced material, and provide enough information for someone else to reproduce your calculations. Do not invent citations or data. The same marking scheme applies to both pathways. **Evidence of understanding:** Code output or a correct number alone is not sufficient for full credit. Interpret the specific numbers and figures you obtain, explain why they make physical sense (or do not), and connect any literature claims to identifiable sources. If you use AI, critically check its answers; AI-generated references are not evidence.

The Imaging Model

A square object is divided into four equal pixels. Each pixel has an unknown nonnegative *linear attenuation coefficient*, in arbitrary units:

$$X = \begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix}, \qquad \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}.$$

For a monochromatic X-ray beam, an idealized attenuation model is

$$I = I_0 e^{-p}, \qquad p = -\ln\left(\frac{I}{I_0}\right) = \sum_{j=1}^{4} \ell_j x_j = \boldsymbol{\ell} \cdot \boldsymbol{x},$$

where I_0 is the incident intensity, I is the measured intensity, and ℓ_j is the beam's path length through pixel j. We use normalized path lengths, and assume each measurement is accurate unless otherwise stated. **The measured quantity p is a logarithmic attenuation, not raw intensity.**

CT in everyday language. Think of each pixel as a small patch of material. A larger attenuation coefficient means that this particular patch weakens the X-ray beam more strongly. A ray crossing two patches collects contributions from both; a ray missing a patch contributes nothing from it. Thus a path-length vector such as $(1, 1, 0, 0)$ means "pass through the top-left and top-right pixels, but not the bottom two." The detector does not directly read the four pixel values; it reports one combined measurement.
Why the logarithm? Real detectors measure X-ray intensity I. The expression $p = -\ln(I/I_0)$ converts the measured intensity ratio into a quantity that, under our simplified assumptions, is a *sum* of contributions from the pixels. This is why linear algebra can help. You do not need prior knowledge of CT or exponential attenuation to complete Parts A–E; use the equations provided below.

The scanner provides the following idealized measurements:

Measurement	Path-length vector $\boldsymbol{\ell}_i$	Measured p_i
1	$(1, 1, 0, 0)$	5
2	$(0, 0, 1, 1)$	9
3	$(1, 0, 1, 0)$	6
4	$(1, 0, 0, 1)$	7

These rays are an instructional abstraction, not a complete model of clinical scanner geometry. For example, measurement 1 adds the two top-row pixel contributions; measurement 2 adds the two bottom-row contributions. Some supplied paths are mathematical simplifications rather than rays a real scanner could trace.

Interpretation guidance: When asked to explain a result, use the pattern *"I found [specific result]. In this model that means [physical interpretation], because [mathematical reason]."* A short, precise explanation is better than a long general paragraph.

Part A: What Is Hidden? (3 marks)

A1. Explain what $x_1, \ldots, x_4$ represent and why they should be nonnegative. Explain what a larger value would mean for the material in that pixel. **[1 mark]**

A2. Explain why an X-ray measurement can be represented as a dot product $\boldsymbol{\ell}_i \cdot \boldsymbol{x}$. Identify the physical meaning of each factor, and explain what a zero entry in a path-length vector means. **[2 marks]**

Part B: Dot Products and Projection (4 marks)

B1. Write the four measurement equations explicitly using the dot products and the supplied data. Choose one equation and explain in plain language which pixels contribute to that measurement. **[2 marks]**

B2. Consider the first measurement direction $\boldsymbol{\ell}_1 = (1, 1, 0, 0)^T$. For a proposed image $\boldsymbol{z} = (1, 4, 2, 5)^T$, calculate the **vector projection** of $\boldsymbol{z}$ onto $\boldsymbol{\ell}_1$. Explain in one sentence why this projection contains information only about the part of $\boldsymbol{z}$ aligned with that measurement direction. Explain whether this one measurement can determine all four pixel values, and why. **[2 marks]**

Part C: Reconstruct by Hand (5 marks)

C1. Construct the coefficient matrix A and measurement vector $\boldsymbol{b}$ such that $A\boldsymbol{x} = \boldsymbol{b}$. State their dimensions. **[2 marks]**

C2. Form the augmented matrix $[A \mid \boldsymbol{b}]$ and use **elementary row operations** to obtain its RREF. Show enough intermediate work to make your method clear. State the reconstructed image X, and briefly explain how you know the four values are uniquely determined. **[3 marks]**

Part D: Reconstruct with Python (4 marks)

D1. Create A and $\boldsymbol{b}$ as NumPy arrays. Use `np.linalg.solve(A, b)` to reconstruct $\boldsymbol{x}$; reshape the result into a 2×2 image and display it using a grayscale `matplotlib` image plot with a labelled colour bar.

Identify which pixel has the greatest attenuation and what that means physically. **[2 marks]**

D2. Verify the result by computing $A\boldsymbol{x}$ in Python and checking agreement with $\boldsymbol{b}$ using `np.allclose`. Explain what this check establishes, and one thing it *does not* establish about a real CT image. Refer to your actual computed values, not only the function name. **[2 marks]**

Python hint: displaying a reconstructed image

We have not used `imshow()` in class. The Matplotlib function `plt.imshow()` displays a two-dimensional NumPy array as an image. You may adapt this example; you do *not* need to memorize the syntax.

```
import numpy as np
import matplotlib.pyplot as plt
pixels = np.array([[2, 3], [4, 5]])
plt.imshow(pixels, cmap="gray", interpolation="nearest")
plt.colorbar(label="Attenuation coefficient")
plt.show()
```

For Part D, reshape your reconstructed vector into a 2×2 array using `x.reshape(2, 2)` before plotting. `cmap="gray"` sets the grayscale colour map, and `interpolation="nearest"` keeps pixel boundaries sharp. This type of code will *not* be tested on an exam, but very useful to learn!

Part E: When Measurements Are Missing (3 marks)

E1. Remove measurement 4. Write the resulting 3×4 augmented matrix and determine its RREF. State the number of free variables and whether the image is uniquely reconstructable. Explain what a free variable means for this image. **[2 marks]**

E2. Give **two different** nonnegative images consistent with the remaining three measurements, and explain what this demonstrates about the available information. Explain why both images can fit the same measured data. **[1 mark]**

Part F: Research Extension: When Reconstruction Goes Wrong (6 marks)

Research question: A mathematical reconstruction may exist, yet be non-unique, unstable, or physically misleading. Choose **one** of the queries below. You must consult **at least two credible external sources**, including **at least one scientific paper, scholarly textbook, or authoritative technical publication**. Explain how the sources informed your model. At least one source must support a specific physical claim you make, and at least one source must help justify a modelling choice or parameter. Cite the relevant source next to the claim, not only in a reference list. AI responses are *not* substitutes for these sources. You may use peer-reviewed articles, scholarly textbooks, or authoritative institutional resources (e.g., medical imaging societies, universities, or national laboratories); do not rely on unsourced websites.

Option 1: Sparse-view imaging (recommended). Research why limited X-ray views create ambiguity or artifacts. Extend the model to a 3×3 image or larger, devise a set of physically interpretable ray-path vectors, and use RREF/rank or numerical experiments to investigate whether the measurements uniquely determine the image. Propose and test at least one additional measurement or justified physical constraint. Explain what remains uncertain.
Option 2: Measurement noise. Research one source of noise or uncertainty in CT measurements. Create a justified noisy version of your measurement vector (state its magnitude and units), reconstruct the image, and quantify the change relative to the original reconstruction. Repeat for at least two noise levels or patterns. Explain why an exact linear-algebra solution does not guarantee physical accuracy.
Option 3: Limits of the attenuation model. Research a real limitation of the monochromatic Beer–Lambert model (for example, beam hardening). State a more realistic equation or documented approximation, then design a small numerical experiment demonstrating the difference between the idealized and modified measurements. Explain which part of your original linear model becomes unreliable.

Required deliverables (all options):

F1. Research and question (1 mark): State your chosen question and summarize relevant findings from at least two credible sources with complete references and a sentence explaining exactly which claim each source supports. **[1 mark]**

F2. Model and assumptions (2 marks): Introduce your mathematical model, define its variables, and justify at least one new assumption or parameter using the literature. Clearly distinguish published information from your own simplifications. Explain why your chosen modification is relevant to a real scanner. **[2 marks]**

F3. Computational investigation (2 marks): Perform a reproducible numerical experiment in Python, with at least one informative figure or table and a meaningful comparison. Report at least one numerical measure of the difference and interpret it in words. **[2 marks]**

F4. Interpretation (1 mark): Explain what you learned, identify an important limitation, and propose one next step to improve the model. State which conclusion is supported by your experiment and which remains uncertain. **[1 mark]**

Scope: You may read about advanced reconstruction methods, but you are *not required* to implement singular value decomposition, filtered backprojection, optimization, or other methods beyond Topics 1–9. A carefully justified, simpler investigation is fully acceptable. The challenge is in researching and designing the model, not in using the most sophisticated code.

Before You Submit

- ☐ Our `.ipynb` file includes all group members' names and only one (1) group member has submitted on behalf of the group.
- ☐ All Parts have been completed.
- ☐ Our mathematical work is clearly presented.
- ☐ All required Python code and outputs are included.

- ☐ All code has sufficient commenting
- ☐ All notebook cells have been run successfully.
- ☐ If GenAI was used, we identified the platform(s) and attached/exported the complete AI conversation(s).
- ☐ Our final `.ipynb` file is uploaded to LEARN Dropbox.

AI Documentation

If you used generative AI at any point:

- state the GenAI platform(s) used; and
- attach/export your **complete conversation(s)** showing your prompts and the responses you received.

Your AI conversation is supporting evidence of your process. You are still responsible for the accuracy and interpretation of anything you use in your submitted notebook.

If you did not use generative AI, state:

We completed this assignment without using generative AI.